EENG 4640 Sustainability Presentation

Project Glass House

Team No. 8

Tamera Gilmore

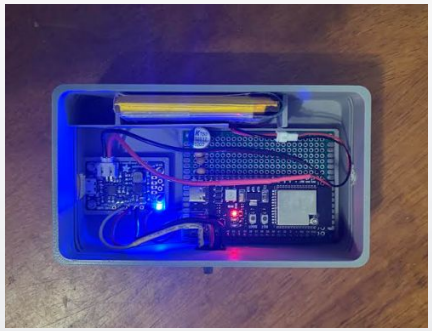
Jack Jones

Bryan Bosely

Cole Freeman

Project Introduction

- Search and rescue operations in collapsed buildings and debris



Shouter node

- Our system uses:
  - 4 ESP32 RF shouting nodes
  - Wi-Fi Channel State Information (CSI)
  - Raspberry Pi processing + LCD Display + 1 sensing node

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Main Issues Related to Sustainability Identified in your project

Sustainability Area	**Focus**
Environmental impact	Proper Disposal of Batteries and Battery Leakage
Energy efficiency	Ultra-low power sensing
Material and Pollution Issues	Recycling and Pollution
Safety Issues	RF safety
Social Issues	Improper Use
Economical Issues	<$450 per system

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Engineering Standards & Compliance

- IEEE 802.11 Wi-Fi standard: 2.4 GHz ISM band [2]
- IEEE C95.1-2019/FCC Title 47: Safety standards for RF exposure; ensuring 2.4 GHz signal density is safe for trapped survivors & central device users [2], [3].
- IEEE Code of Ethics: public safety & welfare [1]

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Environmental Issues - Disposal

Improper Disposal : When the LiPo battery is replaced at end-of-life, disposal in regular trash can cause landfill fires and release harmful chemicals into the environment.

Mitigation: All spent batteries can be removed and sent to certified e-waste/battery recycling programs to safely recover materials and prevent environmental contamination.

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Environmental Issues - Battery Leak

Toxicity: Analysis of lithium-ion (LiPo) chemistry. If enclosures are breached in rubble, electrolyte leakage can contaminate soil.

Mitigation: Use of enclosures to prevent chemical leakage and protect the 2.4 GHz high-gain antennas from breakage.

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Energy Use Issues Central Hub

- Raspberry Pi 4 Dominates Power Budget
- Power Bank Efficiency Loss
- Hub Has No Battery Backup Transition
- Single Point of Failure

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Energy Use Issues Nodes

- Battery Runtime is Marginal for Extended Operations [4]
- Depth-of-Discharge Degradation [5]
- No Runtime Monitoring
- Single Battery, No Redundancy

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Energy Use Advantages

- Distributed peripheral nodes reduce per-device energy footprint [6]
- Battery-powered ESP32s operate autonomously without grid dependency [4]
- Lower carbon footprint than cloud-based or smartphone-dependent systems [6]

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Energy Use Advantages

Device Type	Power Draw	Runtime	Annual Energy
ESP32 Peripheral	0.396W	22 hours (battery)	~3.5 kWh/year
Raspberry Pi 4	4.0W	23 hours	~35 kWh/year
Traditional IoT Hub (cloud)	10-50W	Continuous	~88-438 kWh/year
Smartphone alternative	3-5W	1-2 days	~5-10 kWh/year

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Renewable Energy Use

- Solar Supplementation: To achieve energy neutrality for the sensing nodes, a small 5V/1W solar panel could be integrated.

- A 1W panel in 5 hours of peak sun provides 5Wh. The solar panel would cover 79% of the daily requirement, reducing the “charging cycles” needed.

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Use of Materials Issues

- Antennas: Use of standard 2.4 GHz dipole antennas with recyclable copper elements and plastic housings.

- Enclosures: 3D-printed using PLA, which is recylable and requires less energy to manufacture than traditional injection-molded ABS plastic.

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Waste of Materials Issues

- Additive Manufacturing: 3D printing the RPi and ESP32 mounts ensures zero-waste production compared to traditional subtractive manufacturing [7], [8].

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Pollution Related Issues

Operational pollution:

- No exhaust emissions
- Minimal RF pollution (low power)

Manufacturing footprint exists but is small due to:

- Low component count
- Long service life.

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Safety Issues - RF Exposure

Estimate RF exposure when person holds central device at 0.2m

Assumptions:

Transmit power: 20 dBm = 100mW

Atenna gain: 2 dBi

Frequency: 2.4 GHz

$$S = \frac{(0.1)(10^{\frac{2}{10}})}{4\pi(0.2)^2} = 0.315\,\frac{W}{m^2}$$

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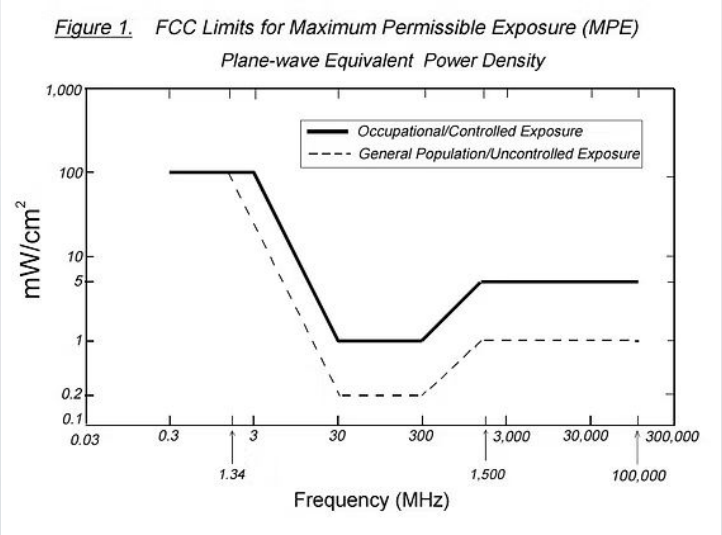
Safety Issues - RF Exposure

- RF Exposure: Calculations show that at a distance of 0.2 meters, the Power Density (S) is about 0.315 W/m^2,

- which is significantly below the IEEE/FCC limit of 10 W/m^2 [3]



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Social Issues

Problem: Could be used to case houses for robberies

Mitigation: Require government issued credentials obtained through licensing agreements with approved organizations

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Constituencies (People Affected)

- Disaster Victims: Gain a "digital voice" even when trapped deep in rubble [8].
- First Responders: Use the RPi map display to prioritize digging locations, reducing exhaustion and risk [9].
- Engineering Team: Responsible for the accuracy of the localization algorithm.

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Economical Issues

- Capital Expense: $450 for a full 4-node + Hub kit.
- Traditional Alternative: Commercial radar systems costing $48,000+ [10], [11].
- Benefit: For the cost of one commercial unit, a city could equip 105 rescue teams with Project Glass House kits.

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People, Planet, Profit Relationship

- People: Enhanced survival rates and rescuer safety.
- Planet: Low-power deterministic algorithms and recyclable enclosures [7].
- Profit: Radical cost-efficiency makes life-saving technology a standard utility rather than a luxury.

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Project Footprint

- Deterministic signal processing, such as the implementation of FFTs, is more efficient and interpretable than machine learning models [12].

- Operational Footprint: The device creates zero air or water pollution and uses existing Wi-Fi spectrum sustainably.

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Conclusions

As engineers, our responsibility is to provide the most reliable and least wasteful tool possible for life-critical missions.

- Economic: $450 vs $48,000+
- Safe: RF exposure well below FCC limits
- Green: zero operational emissions and recyclable 3D printable enclosures
- Ethical: access restricted to prevent misuse

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Questions

- What proactive measures should be taken to ensure that our device is being used ethically?
- How would you redesign the node architecture to maintain coverage if one or more nodes lose power mid-deployment?
- What redundancy strategies would you recommend to eliminate the single point of failure at the central hub?

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